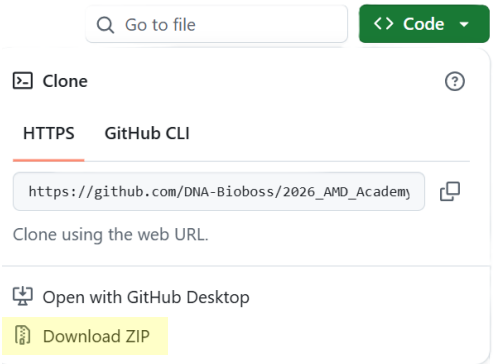
Investigation Background

You work for a state health department that has identified several cases of *Candidozyma auris* (*C. auris*). Your assignment is to explore the investigation data using MicrobeTrace and identify how *C. auris* is spreading in this outbreak. When you are finished email your session to the email address provided below.

MicrobeTrace Instructions:

1. From the GitHub Repository https://github.com/DNA-Bioboss/2026_AMD_Academy_MicrobeTrace_S2.git download and/or clone the data into a folder on your computer
    a. To download, green **"< > Code" > "Download ZIP"**

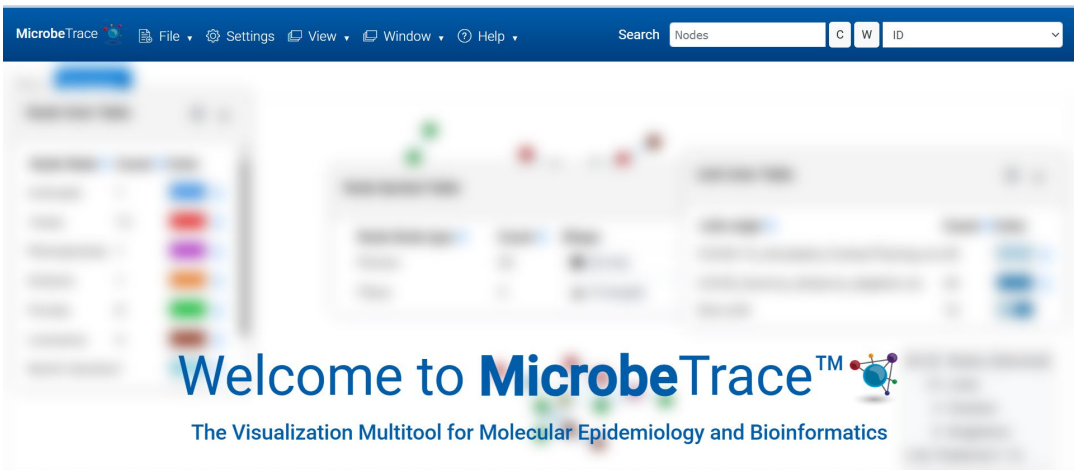


    b. Review the data
        i. **Facility_L** is missing coordinates. Use the following address to find the latitude and longitude: 936 N Andover Rd, Andover, KS 67002
        ii. Directions: Paste the address into google.com/maps, right-click the red pin and when the coordinates appear, click on them to copy them onto your clipboard and paste them onto the excel sheet.
        (link directly to the address in google maps here: 936 N Andover Rd, Andover, KS 67002)

2. In your web browser, open a tab to **MicrobeTrace**

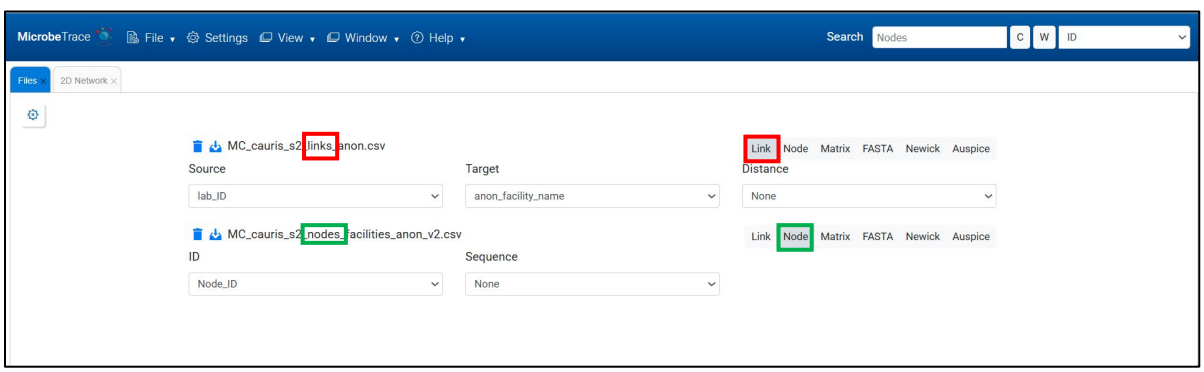
    https://microbetrace.cdc.gov/MicrobeTrace/

    [MicrobeTrace is capable of running on any computer running Google Chrome, Firefox, or Microsoft Edge.]

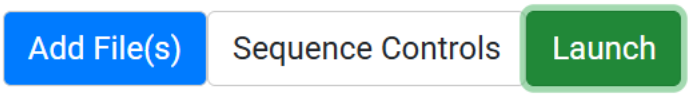


3. Load your updated ***MC_cauris_s2_nodes_facilities_anon_v2.csv*** and ***MC_cauris_s2_links_anon.csv*** files to MicrobeTrace by dropping them into the homepage.

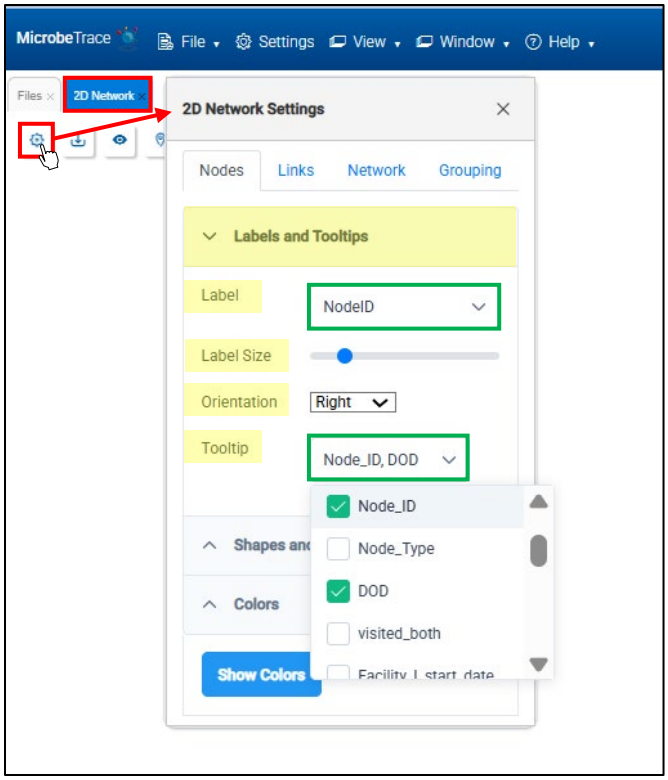
4. A file window will appear with tabs that categorize the file types loaded. Verify that the correct tab is selected for each file.
    a. If the wrong tab is selected, choose the correct tab by clicking on it.



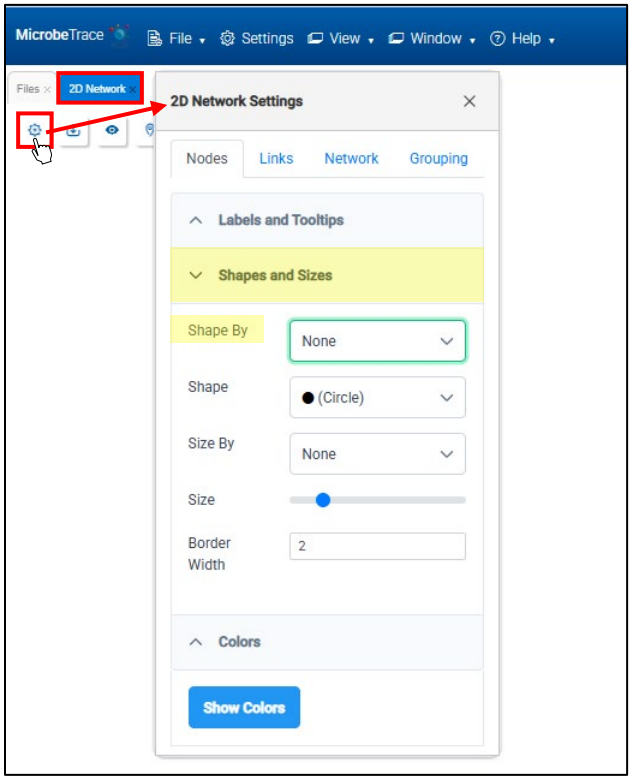
5. Click Launch



6. To modify and add labels to the nodes in the **2D Transmission Network**, click **"Labels and Tooltips"** dropdown in the **2D Network Settings** menu. Select **"NodeID"** and see the labels appear. Use **"Label Size"** and **"Orientation"** to adjust the labels. **"Tooltip"**, allows you to add additional information to a node. This information appears in a small pop-up window when you hover over the node. Under **"Tooltip"** unselect **"_id"** and select **"NodeID"** and **"DOD"** (diagnosis date).



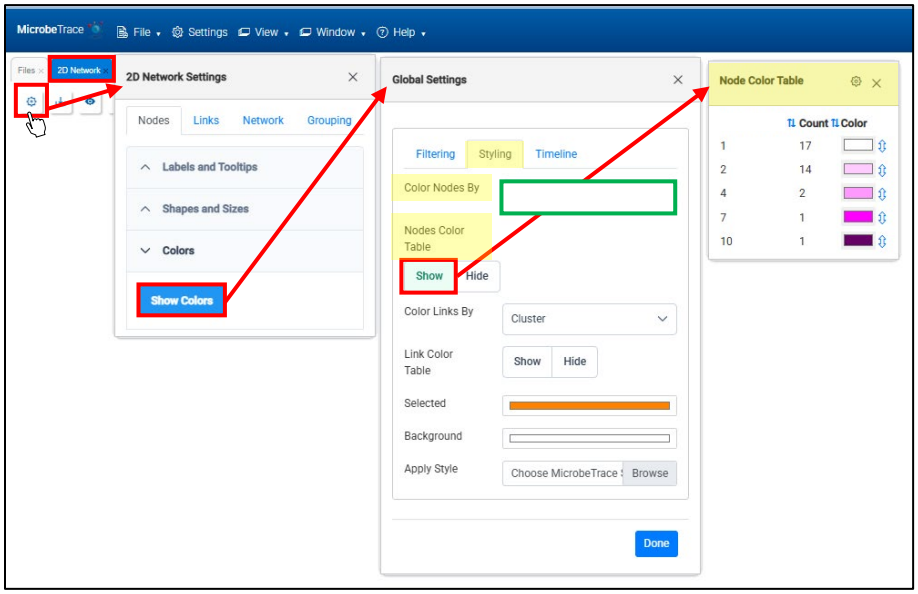
7. Use the **"Shapes and Sizes"** dropdown menu to adjust the shape and size of the nodes.
    a. Under **"Shape By"**, what column should be chosen to distinguish between facilities and cases?



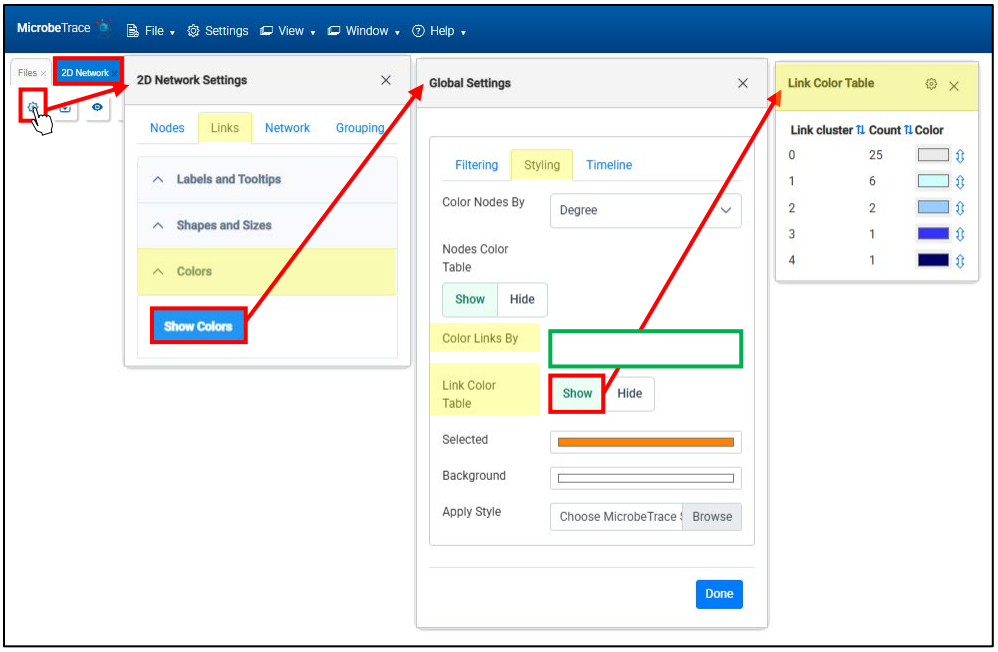
8. Use the **“Node Color Table”** to change the node colors and format the legend.

   [Note: the **Node Color Table** may appear under the **Global Settings Table,** simply drag them to move them on screen]

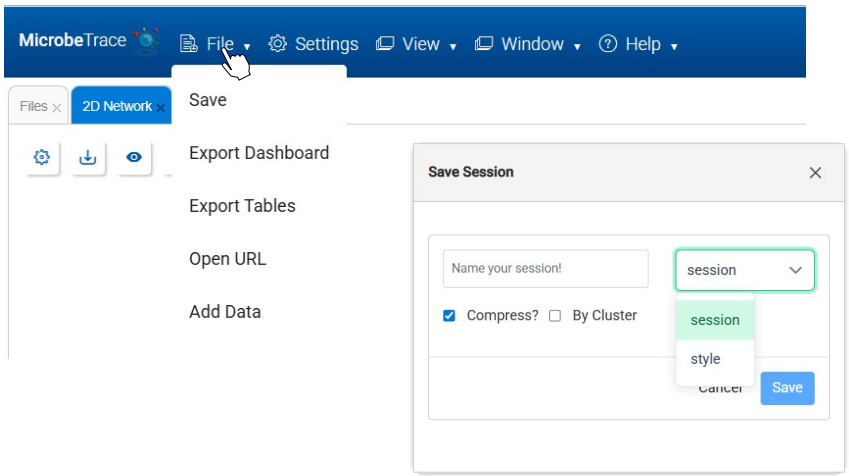
   Color nodes by “visited_both” It will help distinguish which cases visited the facilities of interest.



9. Click the cog icon and select the **“Links”** tab to adjust the link properties as needed.

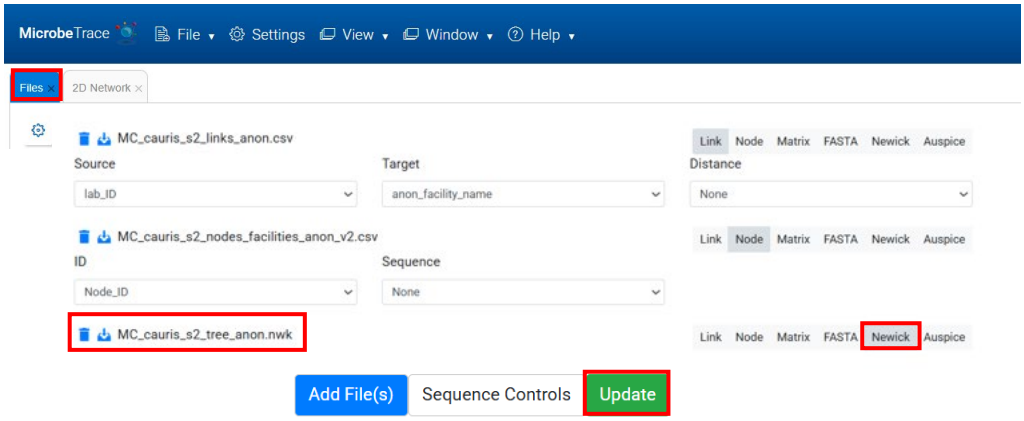


10. <u>Before</u> adding genetic data to the transmission network, save your work and the style of your node and link properties.
    a. Save your session: **Click File > Save**, name your session, verify **"session"** is selected in the dropdown menu and click **"Save"**.
        i. Saving your work lets you save your dataset and come back to it later.

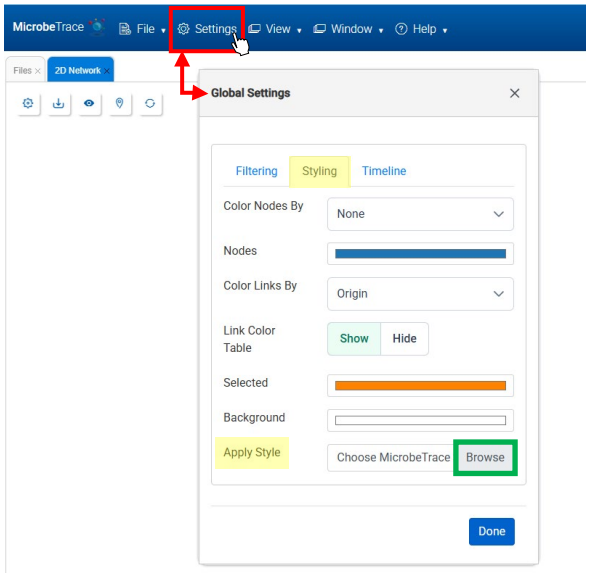


    b. Save your style: **Click File > Save**, name your style, select **"style"** from the dropdown menu, and click Save.
        i. This will allow you to apply the same settings when loading data for another investigation. Keep in mind to use the same file types (i.e., node vs link) with the same variable headers when you apply the style file in your new session, or the saved styles will not apply correctly.

11. Load the genomic data onto MicrobeTrace.
    a. Select the **"Files" tab**, select **"Add File(s)"**, load the ***MC_cauris_s2_tree_anon.nwk*** file, verify that the **"Newick"** tab is selected, and then select **"Update"**.



12. Using the **"Global Settings"** window, apply the style saved in step 10.
    a. Click **"Settings"** under the **"Styling"** tab, scroll down to **"Apply Style"**, select **"Browse"**, and select your saved style file.

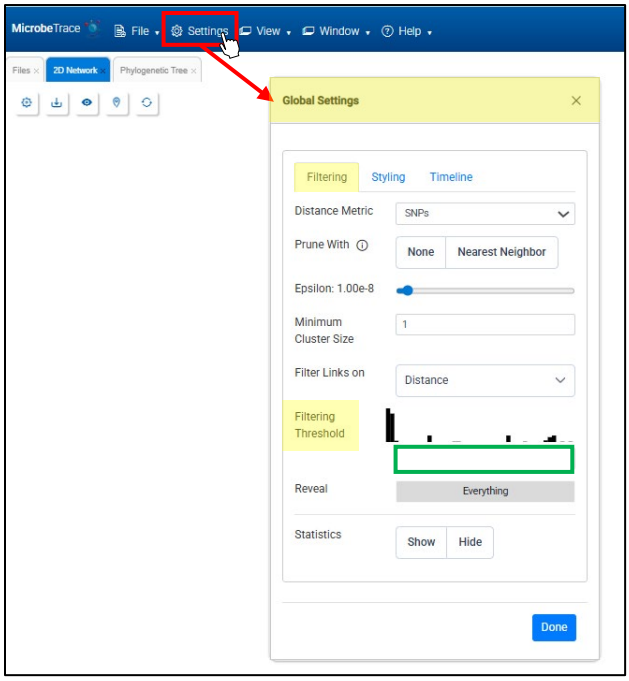


    b. After setting node and link properties, adding genomic data will reset them, removing your customizations. Simultaneously loading epidemiological and genomic data to MicrobeTrace can also make it harder to clearly visualize and adjust node and link properties. For this reason, it is recommended to first load

epidemiological data, customize the node and link properties and save your style. Then add the genomic data and apply the saved style.

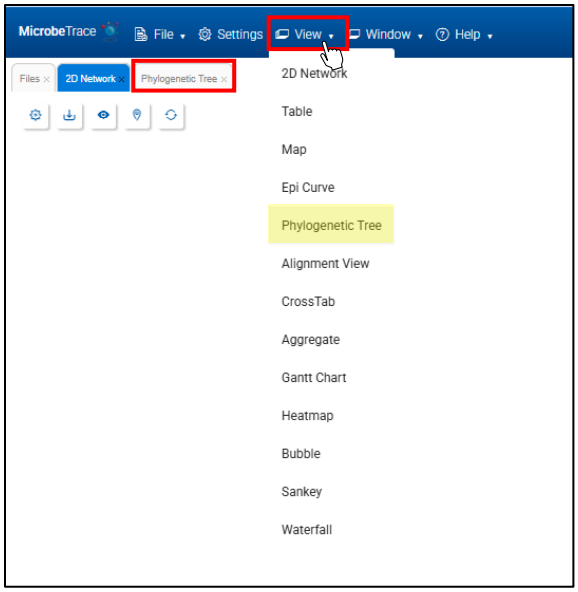
13. Move the nodes around to improve the visualization of the **transmission network**.

14. Change the link threshold in **"Global Settings"**
    a. Select **"Settings" > "Filtering"** tab, under **"Filtering Threshold"**
    b. Refer to the article ***Identifying Candida auris transmission in a hospital outbreak investigation using whole-genome sequencing and SNP phylogenetic analysis*** that was downloaded for the link threshold
        i. HINT: Review the abstract



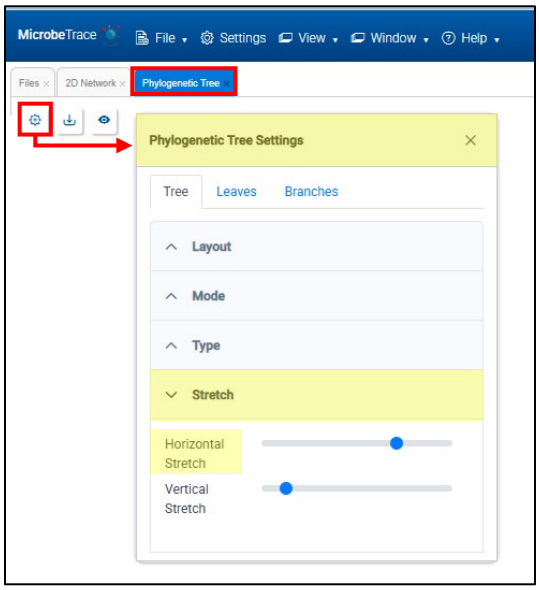
    c. Make sure that "Nearest Neighbor" is selected under "Prune With."

15. View the phylogenetic tree
    a. **View > Phylogenetic Tree**

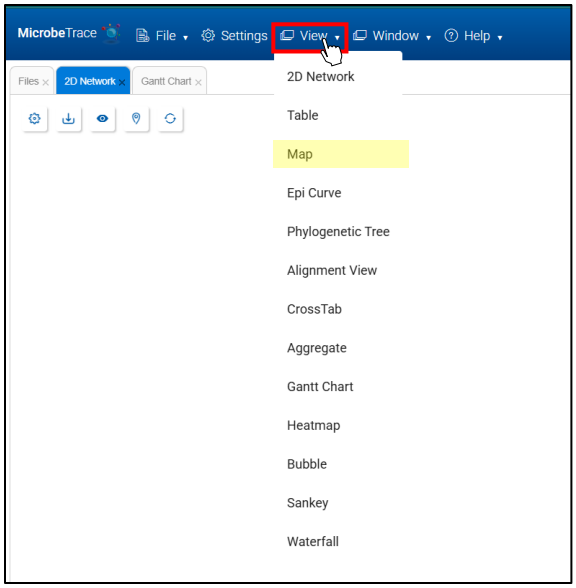


b. The bottom part of the phylogenetic part of the tree is a bit condensed, use the **Horizontal Stretch** to make it easier to view

i. **Cog icon** **> Phylogenetic Tree Settings > Tree > Slide the Horizontal Stretch**

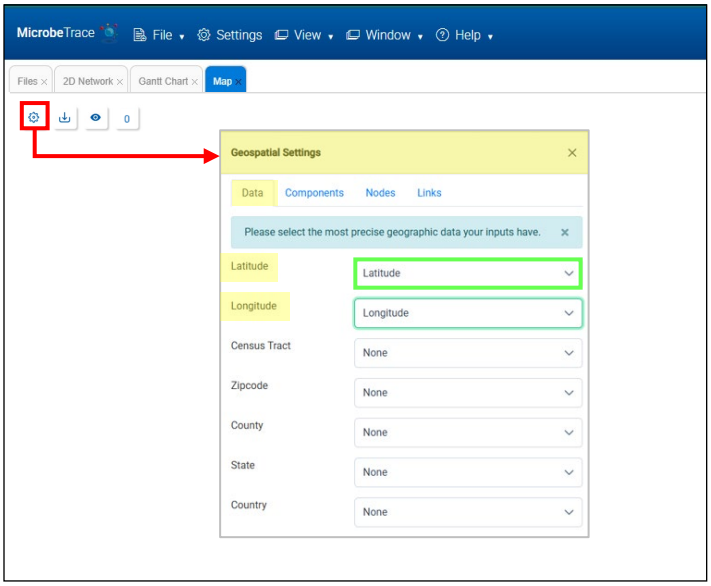


16. View the **Map**



a. Assign **"Latitude"** and **"Longitude"** to their respective columns

b. **Cog Icon** **> Geospatial Settings** menu**> Data**



c. What metropolitan cities are affected?

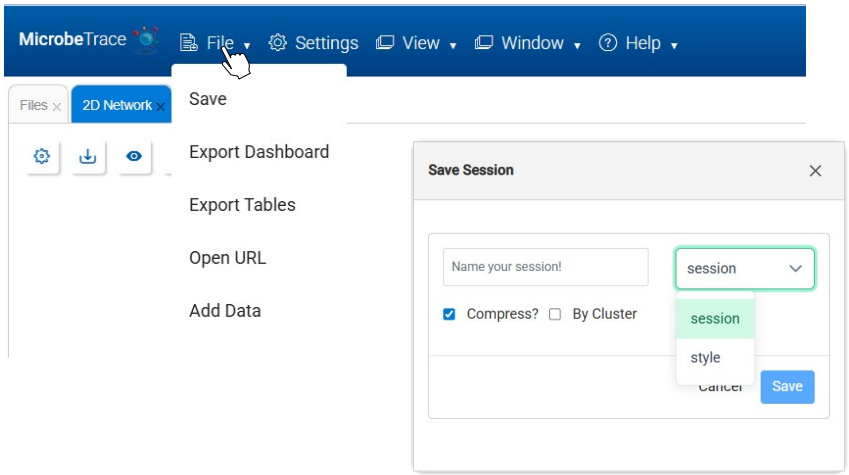
17. Create a view with multiple tabs
    a. Select your **"2D Network"** tab, drag the **"Phylogenetic tree"** onto your network, let go when you see the split screen you'd like.

    b. Now drag the **"Map"** below the phylogenetic tree

18. Write a conclusion that you would write to your HAI/AR epidemiologists
    a. What is occurring based on the MicrobeTrace visualizations?

    b. What are some public health actions you'd recommend?

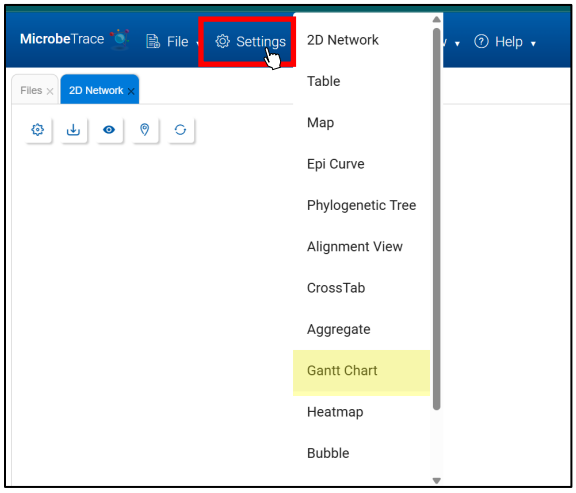
19. Submit your work
    a. **Email your session and summary** to mayra.trujillo@ks.gov



    b. **Remember to include your and your partner's name.**

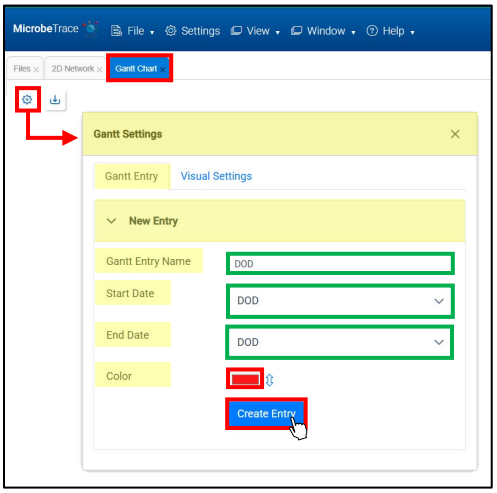
Appendix: Instructions for working with Gantt Charts

1. View the **Gantt Chart**



c. Add the **date of diagnosis (DOD)** in red

   i. **Cog icon** **> Gantt Entry > change parameters > Create Entry**
      1. Type **DOD** under **Gantt Entry Name**
      2. **Start Date**: from the drop down menu select **DOD**
      3. **End Date**: from the drop down menu select **DOD**
      4. Color: Click the color patch and select a red color
      5. Select **"Create Entry"**



d. Add the dates for the two facilities that appear to have the most cases identified in **Step 7**

e. Which cases have overlapping stays?